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SOLUTION FOR EXAMINATION PAPER

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Examination Period :	Mock
Module Code :	MA3508/MA4508/MAT508
Examination Paper Title :	Time series
Duration :	2 hours

Question 1

(i) State the Mercer expansion and the Karhunen-Loeve expansion.

Let x_t be a stochastic process defined on the interval $[a, b]$ such that $\mathbf{E}x_t = 0$ and the covariance function $R(t, s) = \mathbf{E}x_t x_s$ is continuous. Then there is the **Mercer expansion** of the covariance function

$$R(t, s) = \sum_{j=1}^{\infty} \lambda_j \phi_j(t) \phi_j(s),$$

where $\phi_1(t), \phi_2(t), \ldots$ yield an orthonormal basis of $L_2([a, b])$.
Also, there is the **Karhunen-Loeve expansion** of the stochastic process

$$x_t = \sum_{j=1}^{\infty} \epsilon_j \sqrt{\lambda_j} \phi_j(t),$$

where $\{\epsilon_j\}_{j=1,2,\ldots}$ is white noise with unit variance.

[2]

(ii) Show that the correlation function $\rho(t)$ of a stationary process is positive semidefinite.

We need to show that the correlation function $\rho(t)$ of a stationary process x_t satisfies

$$\sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j \rho(t_i - t_j) \geq 0$$

for all n, $t_1, \ldots, t_n$ and $c_1, \ldots, c_n$. Since $R(0)$ is positive, multiplying the inequality on $R(0)$, we obtain

$$\sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j R(t_i - t_j) \geq 0$$

which is needed to prove. Indeed, consider the random variable $\xi = \sum_{i=1}^{n} c_i x_{t_i}$. Then

$$\mathrm{Var}(\xi) = \mathrm{Cov}(\xi, \xi) = \mathrm{Cov}\Big(\sum_{i=1}^{n} c_i x_{t_i}, \sum_{j=1}^{n} c_j x_{t_j}\Big)$$

$$= \sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j \mathrm{Cov}(x_{t_i}, x_{t_j}) = \sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j R(t_i - t_j).$$

Since $\mathrm{Var}(\xi) \geq 0$, we obtain the required inequality.

[3]

(iii) Write the R code for computing the 4-step ahead forecast with its 95% confidence interval for the time series 3.2,1.3,4.5,2.7 from a stationary in wide sense process with the covariance function $R(k) = 1.6e^{-0.7|k|}$.

```
x=c(3.2,1.3,4.5,2.7); xpred=rep(0,4); xsig=rep(0,4);
givenCov=function(k) { return(1.6*exp(-0.7*abs(k))); }
n=length(x); mu=mean(x); y=x-mu;
Gamma=matrix(0,n,n);
for(i in (1:n)) for(j in (1:n))   Gamma[i,j]=givenCov(i-j);
h=4
bh=rep(0,n);   for(i in (1:n))      bh[i]=givenCov(n+h-i);
a=solve(Gamma) %*% bh;
ynh=y%*%a; xpred[h]=mu+ynh;
xsig[h]=sqrt(givenCov(0)-bh %*% solve(Gamma) %*% bh);
confInterval=xpred[h]+xsig[h]*qnorm(1-0.05/2)*c(-1,1)
```

[3]

(iv) Let a process x_t satisfy $x_t - 1.3x_{t-1} = \epsilon_t + 0.8\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Identify if x_t is invertible or not. Give a reason for your answer.

For the given equation, the polynomial $b(z)$ is given by $b(z) = 1 + 0.8z$. The root of $b(z)$ equals $-1/0.8$, which is located outside the unit circle. This means that the given process is invertible.

[2]

(v) Consider the causal AR(3) process $x_t = x_{t-1} - 0.5x_{t-2} - 0.2x_{t-3} + \epsilon_t$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Find the coefficients c_0, c_1, c_2, c_3, c_4 of the MA representation $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$ of this AR(3) process.

The given equation can be written as $a(B)x_t = \epsilon_t$, where $a(z) = 1 - z + 0.5z^2 + 0.2z^3$. Since $\frac{1}{1-u} = 1 + u + u^2 + u^3 + u^4 + \ldots$, we have

$$\frac{1}{a(B)} = \frac{1}{1 - [z - 0.5z^2 - 0.2z^3]} =$$

$$= 1 + [z - 0.5z^2 - 0.2z^3] + [z - 0.5z^2 - 0.2z^3]^2 + [z - 0.5z^2 - 0.2z^3]^3 + [z - 0.5z^2 - 0.2z^3]^4 + \ldots$$

$= 1 + z - 0.5z^2 - 0.2z^3 + z^2 - 2z0.5z^2 + 0.5^2z^4 - 2z0.2z^3 + z^3 - 3z^20.5z^2 + z^4 + \ldots =$ $1 + z + 0.5z^2 - 0.2z^3 - 0.65z^4 + \ldots$. Thus, $c_0 = 1$, $c_1 = 1$, $c_2 = 0.5$, $c_3 = -0.2$, $c_4 = -0.65$.

[6]

(vi) Suppose that the average precipitation in Paris in successive months was approximated by the AR(2) process $x_t - 11 = 0.5(x_{t-1} - 11) - 0.4(x_{t-2} - 11) + \epsilon_t$, where ϵ_t is Gaussian white noise with zero mean and variance 0.6. Suppose that the last successive observations are 13, 10, 15, 14. Find the predictions for the next two months, the mean square error of predictors, and the confidence intervals with confidence level $1 - \alpha$.

Let $x_1 = 13,\ x_2 = 10,\ x_3 = 15,\ x_4 = 14$. Predictions are given by

$$\hat{x}_5 = 11 + 0.5(x_4 - 11) - 0.4(x_3 - 11) = 11 + 0.5 * 3 - 0.4 * 4 = 10.9,$$

$$\hat{x}_6 = 11 + 0.5(\hat{x}_5 - 11) - 0.4(x_4 - 11) = 11 + 0.5 * (-0.1) - 0.4 * 3 = 9.75.$$

Since the MSE is computed on the base of the MA representation, we have

$$\frac{1}{1 - [0.5B - 0.4B^2]} = 1 + [0.5B - 0.4B^2] + [0.5B - 0.4B^2]^2 + \ldots = 1 + 0.5B + (0.25 - 0.4)B^2 + \ldots$$

Therefore, $\mathrm{MSE}(\hat{x}_5) = \sigma^2 = 0.6$ and $\mathrm{MSE}(\hat{x}_6) = \sigma^2(1 + c_1^2) = 0.6(1 + 0.5^2) \approx 0.75$. Then the $(1 - \alpha)$-confidence intervals for x_5 and x_6 are

$$(10.9 - z_{1-\alpha/2}\sqrt{0.6}, 10.9 + z_{1-\alpha/2}\sqrt{0.6}),$$

$$(9.75 - z_{1-\alpha/2}\sqrt{0.75}, 9.75 + z_{1-\alpha/2}\sqrt{0.75}),$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$-quantile of the standard normal distribution.

[5]

(vii) Find at least two values of parameter c such that the stochastic process x_t satisfying

$$x_t = -3cx_{t-1} + 2cx_{t-2} + 6c^2x_{t-3} + \varepsilon_t, \quad t = 0, \pm 1, \ldots,$$

is non-stationary, where ϵ_t is a white noise process with $\mathrm{E}[\epsilon_t] = 0$ and $\mathrm{Var}\,[\epsilon_t] = \sigma_\epsilon^2$.

This process is non-stationary if the equation $a(z) = 0$ has a root on the unit circle. In our case, we have $a(z) = 1 + 3cz - 2cz^2 - 6c^2z^3 = (1 + 3cz)(1 - 2cz^2)$ which has three roots $z_1 = -\frac{1}{3c}$, $z_{2,3} = \pm\sqrt{\frac{1}{2c}}$. Thus at least for $c = 1/2$ and $c = 1/3$ we have that one root lies on the unit circle and, consequently, the process x_t is non-stationary.

[4]

Question 2

(i) Give the definitions of an ARMA(p, q) process with zero mean and an ARMA(p, q) process with mean μ.

1] A stochastic process x_t, $t \in \mathbb{Z}$, is called an **autoregressive moving average** process, ARMA(p, q), with zero mean if x_t is stationary in wide sense and if for every t,

$$x_t - a_1x_{t-1} - a_2x_{t-2} - \ldots - a_px_{t-p} = \epsilon_t + b_1\epsilon_{t-1} + \ldots + b_q\epsilon_{t-q},$$

where ϵ_t is a white noise with mean 0 and variance σ^2, $a_p \neq 0$, $b_q \neq 0$ and polynomials $a(z) = 1 - a_1z - a_2z^2 - \ldots - a_pz^p$ and $b(z) = 1 + b_1z + \ldots + b_qz^q$ have no common factors (roots); $p, q \in \{0, 1, 2, 3, \ldots\}$.
2] A stochastic process x_t is called an ARMA(p, q) **process with mean** μ if the process $y_t = x_t - \mu$ is an ARMA(p, q) process with zero mean.

[2]

(ii) State and prove the condition of causality of the AR(2) process in terms of coefficients.

The AR(2) process with $a(z) = 1 - a_1z - a_2z^2$ is causal (i.e. $|z_1| > 1$ and $|z_2| > 1$) **iff**

$$a_2 + a_1 < 1, \quad a_2 - a_1 < 1, \quad |a_2| < 1.$$

To prove this, we note that $a(z) = 1 - a_1z - a_2z^2$ has no roots within the unit circle **iff** $\tilde{a}(u) = u^2 - a_1u - a_2$ has no roots outside the unit circle.
Roots of $\tilde{a}(u)$ are given by

$$u_1 = \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2}, \quad u_2 = \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{2}.$$

Case $|u_1| < 1$ can be rewritten as $-1 < \frac{a_1+\sqrt{a_1^2+4a_2}}{2} < 1$. Then $-2 < a_1 + \sqrt{a_1^2 + 4a_2} < 2$. Thus, or $-2 - a_1 < \sqrt{a_1^2 + 4a_2} < 2 - a_1$. If $a_1^2 + 4a_2 > 0$, then $2 - a_1 > 0$, $a_1^2 + 4a_2 < (2 - a_1)^2$, that is, $2 > a_1$, $a_2 + a_1 < 1$. If $a_1^2 + 4a_2 < 0$, then u_1 is a complex number and $|u_1|^2 < 1$. Then $|a_1 + i\sqrt{|a_1^2 + 4a_2|}|^2 < 2^2$, then $a_1^2 - (a_1^2 + 4a_2) < 4$, thus $a_2 > -1$. Similarly, case $|u_2| < 1$ can be considered.

[4]

(iii) Write the R code for simulating a realization of Gaussian process with the mean function $m(t) = 1.3/(1+t^2)$ and the covariance function $R(t,s) = e^{-|t-s|} + e^{-2|t-s|}$ at points $0.3, 0.7, 0.9, 1.6$.

```
tp=c(0.3,0.7,0.9,1.6);
mt=1.3/(1+tp**2);
n=length(tp);
CovMtp=matrix(0,nrow=n,ncol=n);
for(i in c(1:n))
{ for(j in c(1:n))
{ CovMtp[i,j]=exp(-abs(tp[i]-tp[j]))+exp(-2*abs(tp[i]-tp[j]));
}}
GausP=mt+t(chol(CovMtp))%*%rnorm(n,0,1);
plot(tp,GausP,type="l",pch=19,main="Gaussian process")
lines(tp,mt,col="red")
```

[3]

(iv) Is the covariance function $R(k) = \frac{5}{(1+|k|^3)^{2/7}}$, $k \in \mathbb{Z}$, short-range or long-range dependent? Explain your answer.

Note that the series $\sum_{k=1}^{\infty} R(k)$ converges/diverges if and only if the series $\sum_{k=1}^{\infty} \frac{5}{k^{3\cdot 2/7}}$ converges/diverges because $\lim_{k\to\infty} R(k)/\frac{5}{k^{6/7}} = 1$. From the course on series, we know $\sum_{k=1}^{\infty} \frac{1}{k^{\alpha}}$ diverges if $\alpha \in (0,1]$. Since $3\cdot 2/7 = 6/7 < 1$, we have that $\sum_{k=1}^{\infty} R(k)$ diverges. Therefore, $R(k)$ is long-range dependent.

[2]

(v) Consider the time series $x_1 = 62$, $x_2 = 67$, $x_3 = 63$ and assume that the observations correspond to a stationary normal process with covariance function $R(k) = 2/\sqrt{1+k^2}$. Compute the best linear predictor for the next observation $\hat{x}_4$ and also the prediction bounds for x_4 with confidence level $1-\alpha$.

Direct calculus gives $\hat{\mu} = (x_1+x_2+x_3)/3 = 64$, $y_1 = x_1 - \hat{\mu} = -2$, $y_2 = x_2 - \hat{\mu} = 3$, $y_3 = -1$, $n = 3$, $h = 1$,

$$\Gamma = \begin{pmatrix} R(0) & R(1) & R(2) \\ R(1) & R(0) & R(1) \\ R(2) & R(1) & R(0) \end{pmatrix} = \begin{pmatrix} 2 & 2/\sqrt{2} & 2/\sqrt{5} \\ 2/\sqrt{2} & 2 & 2/\sqrt{2} \\ 2/\sqrt{5} & 2/\sqrt{2} & 2 \end{pmatrix}, \; b_1 = \begin{pmatrix} R(3) \\ R(2) \\ R(1) \end{pmatrix} = \begin{pmatrix} 2/\sqrt{10} \\ 2/\sqrt{5} \\ 2/\sqrt{2} \end{pmatrix},$$

$$\Gamma^{-1} = \begin{pmatrix} 1.01 & -0.79 & 0.11 \\ -0.79 & 1.62 & -0.79 \\ 0.11 & -0.79 & 1.01 \end{pmatrix}, \quad a = \Gamma^{-1} b_1 = \begin{pmatrix} 0.08 \\ -0.17 \\ 0.79 \end{pmatrix},$$

Then $\hat{y}_4 = a^T Y = -1.47$, and $\hat{x}_4 = \hat{\mu} + \hat{y}_4 = 64 - 1.47 = 62.53$. The MSE of $\hat{x}_4$ is given by $\mathrm{MSE}(\hat{x}_4) = R(0) - b_h^T \Gamma^{-1} b_h = 2 - b_h^T a = 0.99^2$. Then the prediction bounds for x_4 is

$$(62.53 - z_{1-\alpha/2}0.99, 62.53 + z_{1-\alpha/2}0.99).$$

[6]

(vi) A shop sells books. Sales of books (in units of 100) over the past 12 weeks are $x_1, \ldots, x_{12}$

with $x_{12} = 6.2$. The shop believes that the AR(1) process $x_t - 5 = 0.7(x_{t-1} - 5) + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is Gaussian white noise with zero mean and variance 1.44, is relevant for forecasting. Forecast sales for weeks 13, 14, 15, and 16 and give an expression for the confidence interval of future sales for each of the weeks with confidence level $1-\alpha$.

The given process is causal since $|a| < 1$ where $a = 0.7$. Then we have
$\hat{x}_{13} = \mu + a(x_{12} - \mu) = 5 + 0.7 * (6.2 - 5) = 5.84$,
$\hat{x}_{14} = \mu + a^2(x_{12} - \mu) = 5 + 0.7^2 * (6.2 - 5) = 5.588$
$\hat{x}_{15} = \mu + a^3(x_{12} - \mu) = 5 + 0.7^3 * (6.2 - 5) = 5.4116$,
$\hat{x}_{16} = \mu + a^4(x_{12} - \mu) = 5 + 0.7^4 * (6.2 - 5) = 5.28812$.
Since $\sigma = \sqrt{1.44} = 1.2$, the $(1-\alpha)$-confidence intervals for x_{13}, x_{14}, x_{15}, x_{16} are
x_{13} : $(5.84 - z_{1-\alpha/2}1.2, 5.84 + z_{1-\alpha/2}1.2)$
x_{14} : $\left(5.588 - z_{1-\alpha/2}1.2\sqrt{\frac{1-0.7^4}{1-0.7^2}}, 5.588 + z_{1-\alpha/2}1.2\sqrt{\frac{1-0.7^4}{1-0.7^2}}\right) = (5.588 \mp z_{1-\alpha/2}1.46)$
x_{15} : $\left(5.4116 - z_{1-\alpha/2}1.2\sqrt{\frac{1-0.7^6}{1-0.7^2}}, 5.4116 + z_{1-\alpha/2}1.2\sqrt{\frac{1-0.7^6}{1-0.7^2}}\right) = (5.4116 \mp z_{1-\alpha/2}1.58)$
x_{16} : $\left(5.28812 - z_{1-\alpha/2}1.2\sqrt{\frac{1-0.7^8}{1-0.7^2}}, 5.28812 + z_{1-\alpha/2}1.2\sqrt{\frac{1-0.7^8}{1-0.7^2}}\right) = ((5.28812 \mp z_{1-\alpha/2}1.63)$

[4]

(vii) Find the covariance function for a discrete weakly stationary process x_t with spectral density $f(\lambda) = 1 + |\lambda|$, $\lambda \in [-\pi, \pi]$.

The covariance function is computed by

$$R(k) = \int_{-\pi}^{\pi} \cos(k\lambda) f(\lambda) d\lambda.$$

Thus, $R(0) = \int_{-\pi}^{\pi}(1 + |\lambda|)d\lambda = 2\pi + \pi^2$ and for $k > 0$ we have
$R(k) = \int_{-\pi}^{\pi} \cos(k\lambda)d\lambda + \int_{-\pi}^{\pi} \cos(k\lambda)|\lambda|d\lambda = 0 + 2\int_0^{\pi} \cos(k\lambda)\lambda d\lambda = 2/k \int_0^{\pi} \lambda d \sin(k\lambda)$
$= 2/k[\lambda \sin(k\lambda)|_0^{\pi} - \int_0^{\pi} \sin(k\lambda)d\lambda] = 2/k[0 + \frac{1}{k}\cos(k\lambda)|_0^{\pi}] = 2/k[0 + \frac{1}{k}((-1)^k - 1)] = \frac{2((-1)^k-1)}{k^2}$.

[4]

Question 3

(i) Give the definition of the n-order stationary process and the definition of the Wiener process.

1] A process is called n-**order stationary** if

$$F_{t_1,\ldots,t_n}(u_1,\ldots,u_n) = F_{t_1+s,\ldots,t_n+s}(u_1,\ldots,u_n)$$

for all $t_1,\ldots,t_n,u_1,\ldots,u_n,s$.
2] A process W_t, $t \in [0,\infty)$ is called a **Brownian motion** (or **Wiener process**) if

- $W_0 = 0$,
- increments have the normal distribution, $W_s - W_t \sim N(0, s-t)$ for all $0 \le t \le s$,
- increments $W_s - W_t$ and $W_v - W_u$ are independent for all $0 \le t < s \le u < v$.

[2]

(ii) Let x_t be a continuous stationary process with zero mean and the covariance function $R(t)$. Suppose that this process was observed at points $t_1, t_2, \ldots, t_n$, where $t_i < t_{i+1}$. Derive the linear predictor of this process at point t with smallest MSE, $t > t_n$.

We consider the **linear estimator** of x_t in the form

$$\hat{x}_t = a_0 + a_1 x_{t_1} + a_2 x_2 + \ldots + a_n x_{t_n}.$$

An estimator $\hat{x}_{n+h}$ is called the **mean square error** (MSE) **predictor** of x_t if it minimizes $\Psi(a_0, \ldots, a_n) = \text{MSE}(\hat{x}_t) = \mathbf{E}(\hat{x}_t - x_t)^2 = \mathbf{E}(a_0 + a_1 x_n + a_2 x_{n-1} + \ldots + a_n x_1 - x_t)^2$ over all $a_0, a_1, \ldots, a_n \in \mathbf{R}$. A solution of the minimization problem can be found as a solution of the system of equations

$$\frac{\partial \Psi}{\partial a_0}(a_0, \ldots, a_n) = 0, \ldots, \frac{\partial \Psi}{\partial a_n}(a_0, \ldots, a_n) = 0.$$

Note that

$$\Psi(a_0, \ldots, a_n) = \mathbf{E}(a_0 + a_1 x_{t_1} + \ldots + a_n x_{t_n} - x_t)(a_0 + a_1 x_{t_1} + \ldots + a_n x_{t_n} - x_t)$$

$$= \mathbf{E}(a_0^2 + 2\sum_{j=1}^{n} a_0 a_j x_{t_j} - 2a_0 x_t + \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j x_{t_i} x_{t_j} - 2\sum_{j=1}^{n} a_j x_{t_j} x_t + x_t^2)$$

$$= a_0^2 + 2\sum_{j=1}^{n} a_0 a_j \mathbf{E}x_{t_j} - 2a_0 \mathbf{E}x_t + \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j \mathbf{E}x_{t_i} x_{t_j} - 2\sum_{j=1}^{n} a_j \mathbf{E}x_{t_j} x_t + \mathbf{E}x_t^2$$

$$= a_0^2 + 2\sum_{j=1}^{n} a_0 a_j \mu - 2a_0\mu + \sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j (R(t_j - t_i) + \mu^2) - 2\sum_{j=1}^{n} a_j (R(t - t_j) + \mu^2) + R(0) + \mu^2$$

where $\mu = \mathbf{E}x_{t_j} = 0$ is the mean of stationary process and $R(t_j - t_i) = \text{Cov}(x_{t_i}, x_{t_j}) = \mathbf{E}x_{t_i} x_{t_j} - \mu^2$ is the covariance function.
A solution of the system of equations

$$\frac{\partial \Psi}{\partial a_0}(a_0, \ldots, a_n) = 0, \ldots, \frac{\partial \Psi}{\partial a_n}(a_0, \ldots, a_n) = 0.$$

is given by $a_0 = 0$ and $a = (a_1, \ldots, a_n)^T = \Gamma^{-1} b_t$ where $b_t = (R(t - t_1), \ldots, R(t - t_{n-1}), R(t - t_n))^T$ and

$$\Gamma = \Big(R(i-j)\Big)_{i,j=1}^{n} = \begin{pmatrix} R(0) & R(t_2 - t_1) & R(t_3 - t_1) & \ldots & R(t_n - t_1) \\ R(t_2 - t_1) & R(0) & R(t_3 - t_2) & \ldots & R(t_n - t_2) \\ R(t_3 - t_1) & R(t_3 - t_2) & R(0) & \ldots & R(t_n - t_3) \\ \vdots & \vdots & \vdots & & \vdots \\ R(t_n - t_1) & R(t_n - t_2) & R(t_n - t_3) & \ldots & R(0) \end{pmatrix}.$$

[5]

(iii) Write the R code for simulating a realization of length 160 from the ARMA(3,1) process defined by the difference equation $x_t - 0.7x_{t-1} + 0.4x_{t-2} - 0.1x_{t-3} = \epsilon_t + 0.5\epsilon_{t-1}$ with $x_1 = 2.1$, $x_2 = -1.2$ and $x_3 = 1.7$, where ϵ_t is white noise with zero mean and variance $\sigma^2 = 1.7$.

```
n=160;
eps=rnorm(n,0,sqrt(1.7));
x=rep(0,n);
x[1]=2.1; x[2]=-1.2; x[3]=1.7;
for(t in (4:n))
x[t]=0.7*x[t-1]-0.4*x[t-2]+0.1*x[t-3]+eps[t]+0.5*eps[t-1];
plot((1:n), x, pch=19, main="ARMA(3,1) process")
lines((1:n),x)
```

[3]

(iv) Let a process x_t satisfy $x_t + \frac{8}{3}x_{t-1} - x_{t-2} = \epsilon_t - \frac{5}{8}\epsilon_{t-1}$, where ϵ_t is white noise, $t \in \mathbb{Z}$. Identify if x_t is causal or not. Give a reason for your answer.

For the given equation, the polynomial $a(z)$ is given by $a(z) = 1 + (8/3)z - z^2 = 1 + (3 - 1/3)z - z^2 = (1 - z/3)(1 + 3z)$. The roots of $a(z)$ are 3 and $1/3$. The second root is not located outside the unit circle. This means that the given process is not causal.

[3]

(v) Consider the time series $x_k = 1.4\sin(6\pi k/23) + 0.6\sin(4\pi k/13) + 2\sin(3\pi k/11)$, $k \in \mathbb{Z}$, and the filter with spectrum $f(\lambda) = 1_{[0.25\pi, 0.3\pi]}(\lambda)$, $\lambda \in (0, \pi)$. Find the filtered series and its spectrum.

The spectrum of x_k can be written as $1.4^2\delta_{6\pi/23}(\lambda) + 0.6^2\delta_{4\pi/13}(\lambda) + 2^2\delta_{3\pi/11}(\lambda)$. The spectrum of the filtered series is the product of the spectrum of x_k and the spectrum of the filter. Since $6\pi/23 \in [0.25\pi, 0.3\pi]$, $3\pi/11 \in [0.25\pi, 0.3\pi]$, $4\pi/13 \notin [0.25\pi, 0.3\pi]$, we have that the spectrum of the filtered series is $1.4^2\delta_{6\pi/23}(\lambda) + 2^2\delta_{3\pi/11}(\lambda)$ and the filtered series is given by $y_k = 1.4\sin(6\pi k/23) + 2\sin(3\pi k/11)$.

[3]

(vi) Suppose that $x_1, \ldots, x_n$ is a time series from an invertible MA(1) process $x_t = \epsilon_t + b_1\epsilon_{t-1}$, where ϵ_t is white noise. Suppose also that the estimator of the mean of the time series is $\bar{X}_n = 2.7$ and the estimator of the covariance function is $\hat{R}(0) = 4.8$ and $\hat{R}(1) = 1.2$. Estimate parameters of this process: the coefficient b_1 and the noise variance $\sigma^2 = \mathrm{Var}(\epsilon_t)$.

First we have $\hat{\rho}(1) = \hat{R}(1)/\hat{R}(0) = 1.2/4.8 = 1/4$. Now we have to solve $\frac{b}{1+b^2} = \hat{\rho}(1) = 1/4$. Rewriting we have $b^2 - 4b + 1 = 0$ which has roots 0.27 and 3.73. Choosing a root whose absolute value is smaller 1, we have $\hat{b} = 0.27$. Now the noise variance is $\hat{\sigma}^2 = \hat{R}(0)/(1 + \hat{b}_1^2) = 4.8/(1 + 0.27^2) = 4.47$.

[4]

(vii) Find the variance and correlation function of the AR(2) process $x_t = 0.8x_{t-1} - 0.4x_{t-2} + \epsilon_t$, $t \in \mathbb{Z}$, where ϵ_t is white noise with zero mean and variance $\sigma^2 = 1.3$, using Yule-Walker equations.

The 1-st and 2-nd Yule-Walker equations are

$$\rho(1) = 0.8 - 0.4\rho(1),$$
$$\rho(2) = 0.8\rho(1) - 0.4.$$

Therefore, $\rho(1) = \frac{0.8}{1+0.4} = 0.8/1.4 = 4/7$ and $\rho(2) = 0.8 * 4/7 - 0.4 = 0.057$. Then from k-th Yule-Walker equation, we have $\rho(k) = 0.8\rho(k-1) - 0.4\rho(k-2)$, $k = 2, 3, \ldots$. In particular, $\rho(3) = 0.8 * 0.057 - 0.4 * 4/7 = -0.183$, $\rho(4) = -0.8 * 0.183 - 0.4 * 0.057 = -0.169$.
The variance is $\sigma_x^2 = R(0) = \mathbf{E}x_t x_t = 0.8^2 R(0) + 0.4^2 R(0) + \sigma^2 + 2(-0.4)0.8R(1) = R(0)(0.64 + 0.16 - 0.64\rho(1)) + \sigma^2 = \sigma_x^2(0.8 - 0.64 * 4/7) + \sigma^2 = 0.434\sigma_x^2 + \sigma^2$. Thus, $\sigma_x^2 = \sigma^2/(1 - 0.434) = 1.3/0.566 = 2.297$

[5]